ADVANCE NLP (AML23702)

LAB EXERCISE: 5

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Experiment Title	Synonyms, Definitions and Word Senses using NLTK WordNet

Aim / Objective

Purpose: *States what the experiment sets out to build and why.*

This exercise builds a Python program that uses NLTK's WordNet interface to list every sense of a given word with its definition, examples and synonyms. It then tells the meanings apart in different contexts with the Lesk word-sense disambiguation algorithm. The program is tested on “bank” (from the provided notebook), “spring” and “bass”, and runs in a notebook and in a web tab.

Expected Outcomes Addressed

Purpose: *Maps the work back to the specific expected outcomes listed in the lab manual for this experiment.*

- **Outcome 1: Retrieve all senses of a word** — Satisfied by wn.synsets(word), e.g. 18 senses for “bank” and 9 for “bass”.
- **Outcome 2: Identify synonyms** — Satisfied by the lemma names of each synset (e.g. fountain, natural spring, outflow, outpouring for spring.n.03) and a combined list of distinct synonyms.
- **Outcome 3: Identify definitions** — Satisfied by synset.definition() and the first usage example of every sense.
- **Outcome 4: Distinguish meanings in context** — Satisfied by nltk.wsd.lesk on each tokenised context sentence.
- **Outcome 5: Check the sense selection** — Satisfied by comparing Lesk's choice with the intended meaning for 7 sentences (5 correct).

Tools & Technologies Used

Purpose: *Gives a quick technical inventory so a reader knows the stack before reading implementation details.*

Category	Tool / Library	Version	Purpose
Language / Runtime	Python	3.13	Core implementation of the lab program and backend
Lexical Database	NLTK WordNet interface (WordNet 3.0)	NLTK 3.10.4	Synsets, lemmas (synonyms), definitions, examples
Word-Sense Disambiguation	nltk.wsd.lesk	NLTK 3.10.4	Chooses the sense whose gloss overlaps most with the sentence
Tokeniser	nltk.word_tokenize (punkt_tab)	NLTK 3.10.4	Tokenising context sentences
Backend Framework	Flask	3.1.3	JSON API, one POST endpoint per lab question (/api/q1 … /api/q5)

Category	Tool / Library	Version	Purpose
Frontend	HTML5 + CSS3 + vanilla JavaScript	—	Single-page workbench with one tab per lab question
Notebook	Jupyter (nbconvert)	7.x	Step-by-step implementation with saved outputs
Screenshot Capture	Playwright + Google Chrome (headless)	—	Real screenshots of the running frontend

Dataset / Corpus Description

Purpose: *Establishes what data the system was built and tested on.*

The lexical data is Princeton WordNet 3.0, distributed with NLTK (about 117,000 synsets), plus the Open Multilingual WordNet (omw-1.4) package. For testing, I wrote seven context sentences for three ambiguous words: two for "bank" (from the provided notebook), three for "spring" (season, metal coil, water source) and two for "bass" (music, fish).

Sample Text Record:

> *"We drank cold water from a mountain spring." → intended sense spring.n.03 (a natural flow of ground water)*

System Architecture / Pipeline Design

Purpose: *Shows the end-to-end flow of the system visually before diving into code.*

```
 Word --> wn.synsets(word) --> for each synset: name, POS, definition(), lemmas(),
examples()
                                                    |
                                                    v
 Context sentences --> word_tokenize --> lesk(tokens, word) --> chosen synset + definition
per sentence
```

- **Module 1 (Input):** The word and one or more context sentences (web tab or notebook).
- **Module 2 (Sense lookup):** wn.synsets(word) returns noun, verb, adjective and adverb senses.
- **Module 3 (Synonyms & definitions):** Lemma names (underscores replaced by spaces) give the synonyms; definition() and examples() describe the sense.
- **Module 4 (Context disambiguation):** Each sentence is tokenised and lesk() returns the sense with the largest gloss overlap.
- **Module 5 (Output):** A sense table, the combined synonym list and a per-sentence table of chosen senses.

Backend Implementation

Purpose: *Explains the core logic that makes the system work (notebooks/Q5_WordNet_Senses.ipynb and nlp_core.q5_wordnet).*

6.1 Approach / Algorithm

- **Sense listing:** Each synset name (e.g. bank.n.01) encodes the lemma, the part of speech and the sense number.
- **Synonyms:** All lemma names of a synset, collected across senses into a sorted set. The query word itself is excluded from the combined list.

- **Simplified Lesk:** For every candidate synset, count the words shared between its gloss and the sentence, and choose the synset with the highest count.
- **Data setup:** NLTK resources are downloaded only if nltk.data.find() cannot locate them.

6.2 Key Code Snippets

1. Synonyms and definitions of every sense (provided notebook code)

```
def explore_word(word):
    synsets = wn.synsets(word)

    if not synsets:
        print("No WordNet entries found for:", word)
        return

    print("Word:", word)
    print("\nPossible meanings:")

    for i, synset in enumerate(synsets, start=1):
        synonyms = sorted({
            lemma.name().replace("_", " ")
            for lemma in synset.lemmas()
        })
        print(f"\n{i}. Synset: {synset.name()}")
        print("   Definition:", synset.definition())
        print("   Synonyms:", ", ".join(synonyms))

explore_word("bank")
```

2. Choosing the meaning that fits each sentence (Lesk)

```
contexts = [
    "Flowers bloom early in the spring season.",
    "The old mattress had a broken metal spring.",
    "We drank cold water from a mountain spring.",
]
for sentence in contexts:
    sense = lesk(nltk.word_tokenize(sentence), "spring")
    print(f"{sentence}\n    -> {sense.name()}: {sense.definition()}\n")
```

3. Backend function used by POST /api/q5 (nlp_core.py)

```
def q5_wordnet(word, contexts):
    senses = []
    for s in wn.synsets(word):
        senses.append({"synset": s.name(), "pos": s.pos(), "definition": s.definition(),
                       "synonyms": sorted({l.name().replace("_", " ") for l in
s.lemmas()}),
                       "examples": s.examples()[:2]})
    all_syn = sorted({x for s in senses for x in s["synonyms"] if x.lower() !=
word.lower()})
    ctx = []
    for sent in contexts:
        sense = lesk(nltk.word_tokenize(sent), word)
        ctx.append({"sentence": sent, "synset": sense.name() if sense else None,
                    "definition": sense.definition() if sense else "No sense selected"})
    return {"word": word, "num_senses": len(senses), "senses": senses, "all_synonyms":
all_syn, "contexts": ctx}
```

Frontend / UI Implementation

Purpose: *Documents the interface used for pipeline execution and output representation.*

7.1 Framework Choice

The interface is a plain HTML, CSS and JavaScript single page (frontend/index.html, style.css, app.js), so it needs no build step. Flask serves it at http://localhost:5000. A dark-green sidebar lists the five lab questions as tabs; this exercise uses the **05 Word Senses** tab. Clicking the button sends the input to the Flask endpoint with fetch(), and the JSON response is drawn as tables and highlighted text.

7.2 Interface Features

- Word box and context-sentence box (one sentence per line), with a note explaining how to read synset names.
- "Meaning in context (Lesk)" table: sentence, chosen synset and its definition.
- Chips listing every distinct synonym of the word.
- Full WordNet sense table: synset, POS, definition with an example, and synonyms.

Sample Input & Output

Purpose: *Gives concrete, reproducible evidence that the system works.*

Input Sentence / Context	Processing Component	Output
Word: "spring"	wn.synsets()	spring.n.01 season, spring.n.02 metal device, spring.n.03 natural flow of ground water, … (noun and verb senses)
Word: "spring", sense spring.n.03	Synonyms	fountain, natural spring, outflow, outpouring, spring
"The old mattress had a broken metal spring."	Lesk	spring.n.02 – a metal elastic device that returns to its shape …
"We caught a large bass while fishing in the lake."	Lesk	bass.n.02 – the lowest part in polyphonic music (wrong)

Frontend evidence — Word Senses tab, live in the browser:

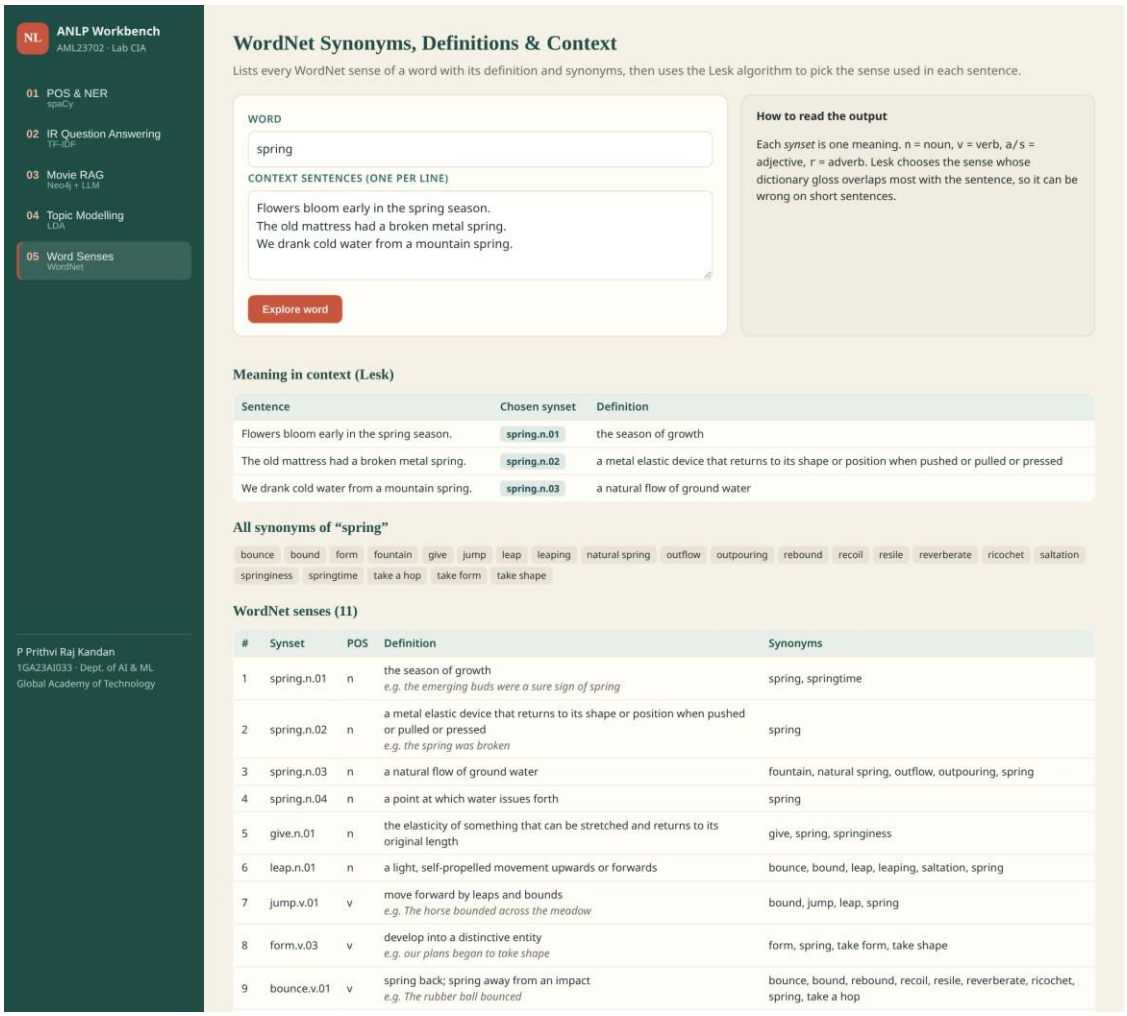


Figure 1. Q5 tab for "spring": sense chosen for each context, all synonyms and the start of the sense table.

Results (Deliverable-Specific Outputs)

Purpose: *Presents the deliverable required by the lab manual: synonyms, definitions and the meaning of a word in context.*

Senses of "bank" (first four of 18)

- **bank.n.01:** sloping land (especially the slope beside a body of water). Synonyms: bank.
- **depository_financial_institution.n.01:** a financial institution that accepts deposits and channels the money into lending activities. Synonyms: banking company, banking concern, depository financial institution.
- **bank.n.03:** a long ridge or pile. Synonyms: bank.
- **bank.n.04:** an arrangement of similar objects in a row or in tiers. Synonyms: bank.

```
Possible meanings:

1. Synset: bank.n.01
   Definition: sloping land (especially the slope beside a body of water)
   Synonyms: bank

2. Synset: depository_financial_institution.n.01
   Definition: a financial institution that accepts deposits and channels the money into lending activities
   Synonyms: bank, banking company, banking concern, depository financial institution

3. Synset: bank.n.03
   Definition: a long ridge or pile
   Synonyms: bank

4. Synset: bank.n.04
   Definition: an arrangement of similar objects in a row or in tiers
   Synonyms: bank

5. Synset: bank.n.05
   Definition: a supply or stock held in reserve for future use (especially in emergencies)
   Synonyms: bank

6. Synset: bank.n.06
   Definition: the funds held by a gambling house or the dealer in some gambling games
   Synonyms: bank

7. Synset: bank.n.07
   Definition: a slope in the turn of a road or track; the outside is higher than the inside in order to reduce the effects o
f centrifugal force
   Synonyms: bank, camber, cant

8. Synset: savings_bank.n.02
   Definition: a container (usually with a slot in the top) for keeping money at home
   Synonyms: bank, coin bank, money box, savings bank
```

Figure 2. Notebook output: senses of "bank" with definitions and synonyms (first entries).

Meaning of a Word in Different Contexts (Lesk)

Sentence	Sense chosen by the program	Correct?
Flowers bloom early in the spring season.	spring.n.01 – the season of growth	Yes
The old mattress had a broken metal spring.	spring.n.02 – a metal elastic device …	Yes
We drank cold water from a mountain spring.	spring.n.03 – a natural flow of ground water	Yes
They sat on the bank of the river.	bank.n.01 – sloping land …	Yes
She deposited her money at the bank.	savings_bank.n.02 – a container for keeping money at home	No
He plays the bass guitar in a rock band.	bass.n.02 – the lowest part in polyphonic music	Yes
We caught a large bass while fishing in the lake.	bass.n.02 – the lowest part in polyphonic music	No

```
In [6]: contexts = [
            "Flowers bloom early in the spring season.",
            "The old mattress had a broken metal spring.",
            "We drank cold water from a mountain spring.",
        ]
        for sentence in contexts:
            sense = lesk(nltk.word_tokenize(sentence), "spring")
            print(f"{sentence}\n   -> {sense.name()}: {sense.definition()}\n")

Flowers bloom early in the spring season.
   -> spring.n.01: the season of growth

The old mattress had a broken metal spring.
   -> spring.n.02: a metal elastic device that returns to its shape or position when pushed or pulled or pressed

We drank cold water from a mountain spring.
   -> spring.n.03: a natural flow of ground water
```

Figure 3. Notebook output: Lesk tells the three senses of "spring" apart.

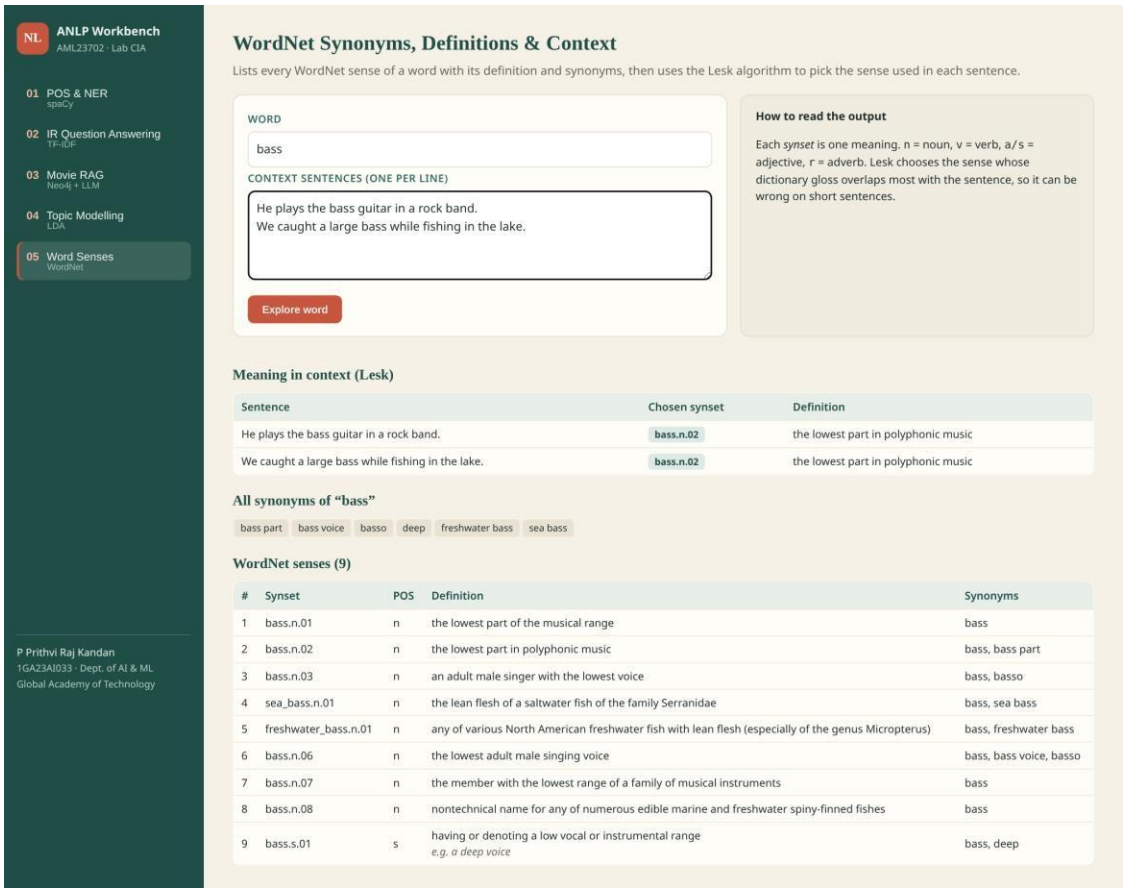


Figure 4. Q5 tab for "bass": the fishing sentence is wrongly given the music sense.

Evaluation / Analysis

Purpose: *Assesses how often the Lesk algorithm picks the intended sense for the 7 test sentences.*

Performance Evaluation

Method	Correct	Accuracy	Remarks
Simplified Lesk (nltk.wsd.lesk)	5/7	71.4%	All three "spring" sentences correct; fails on "deposited money at the bank" and the fishing "bass"

Interpretation

- Lesk succeeds when the sentence repeats a word from the right gloss: season, metal and water for "spring", and river/slope for "bank".
- "She deposited her money at the bank" went to the piggy-bank sense, because that gloss contains "money". The financial-institution gloss uses "deposits", and simplified Lesk does no stemming.
- None of the fish glosses for "bass" contains caught, fishing or lake, so Lesk fell back to a music sense. Extended Lesk, a part-of-speech hint or embedding-based WSD would help.

Challenges Faced & Solutions

Challenge 1: NLTK data download blocked

- **Problem:** nltk.download() could not reach its server from the build machine and printed security errors into the notebook output.
- **Mitigation:** The WordNet, OMW-1.4 and punkt_tab packages were installed from the official nltk_data repository. The notebook now downloads a resource only if nltk.data.find() cannot locate it.

Challenge 2: Many senses with no extra synonyms

- **Problem:** Most senses of "bank" have only "bank" as a lemma, so a synonym-only view looks empty.
- **Mitigation:** Each sense also shows its definition and an example, and a separate list collects every distinct synonym across all senses.

Conclusion

Purpose: *Wraps up project execution and learning outcomes.*

A Python program using NLTK WordNet was built. It lists every sense of a word with its definition, examples and synonyms, and uses the Lesk algorithm to pick the meaning used in a sentence. Lesk chose correctly in 5 of 7 test sentences, including all three senses of "spring". The two failures show its dependence on exact word overlap with short glosses.

Project Repository (GitHub Link)

The complete project (Flask backend, notebooks, HTML/CSS/JS frontend, README and requirements.txt) is uploaded to a GitHub repository. The link is given below.

GitHub Repository URL	https://github.com/vijwalsheshadri/Vijwal1GA23AI057-ANLP
Repository Visibility	Public
Branch / Commit Used for Submission	main

README.md has the setup steps (pip install -r requirements.txt, python -m spacy download en_core_web_sm), the NLTK data download and the run command (python app.py, then open http://localhost:5000).

References

- NLTK Documentation – WordNet Interface: https://www.nltk.org/howto/wordnet.html
- NLTK Documentation – nltk.wsd (Lesk algorithm): https://www.nltk.org/api/nltk.wsd.html
- Miller, G. A. (1995). WordNet: A Lexical Database for English. Communications of the ACM, 38(11), 39–41.
- Lesk, M. (1986). Automatic sense disambiguation using machine readable dictionaries. Proceedings of SIGDOC '86.

ADVANCE NLP (AML23702)
LAB EXERCISE EVALUATION SHEET

Student Name	Vijwal Sheshadri
USN	1GA23AI057
Lab Exercise No.	5
Experiment Title	Synonyms, Definitions and Word Senses using NLTK WordNet

Marks Awarded

Sl. No.	Evaluation Criteria	Max Marks	Marks Obtained
1	Problem Understanding & Scope	5	
2	Pipeline Design & Methodology	10	
3	Implementation Correctness	10	
4	Use of NLP Tools / Models	5	
5	Output Quality / Task Performance	10	
6	Analysis & Interpretation	5	
7	Documentation & Deliverables	2.5	
8	Presentation & Reproducibility	2.5	
	Total	**50**	

Overall Comments

Date of Submission	Signature of Faculty